

TARGET GATE 2026

SUPER 20

SERIES

Compiler Design

CS & IT

YOUR TOP RANK SHORTCUT



By Venkat SIR

(2-6) marks

TOPICS
TO BE COVERED



Phases of Compiler

- (1) Lexical Analysis
- (2) Syntax Analysis (or) Parsing
- (3) Syntax Directed translation (SDT)
- (4) I.C.G
- (5) Optimization

#Q. Which of the following compiler has lexical analysis to identify the tokens of the program?

- ☐ A C compiler
- ☐ B C++ compiler
- ☐ C Java compiler
- ☐ D Operating system

Question

Front end of Compiler

#Q. Which of the following compiler phases depends on Machine?

☒ A Syntax Analysis

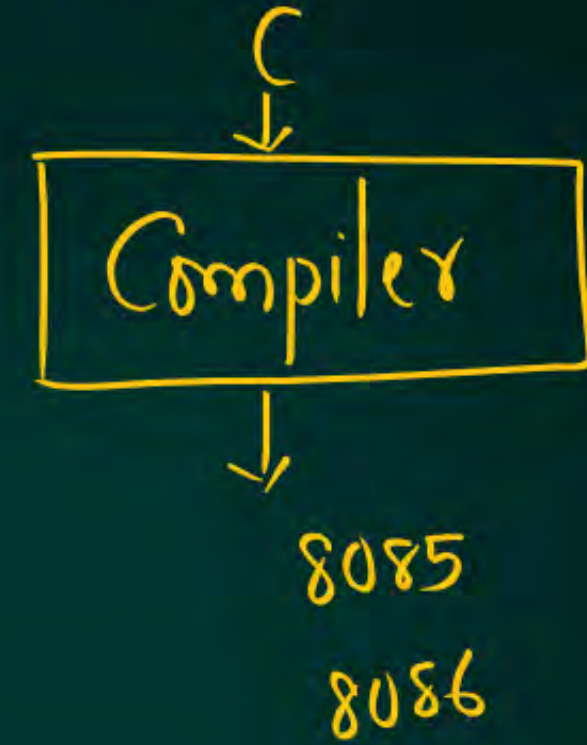
☒ B Intermediate Code Generation

☒ C Code Generator

machine dependent

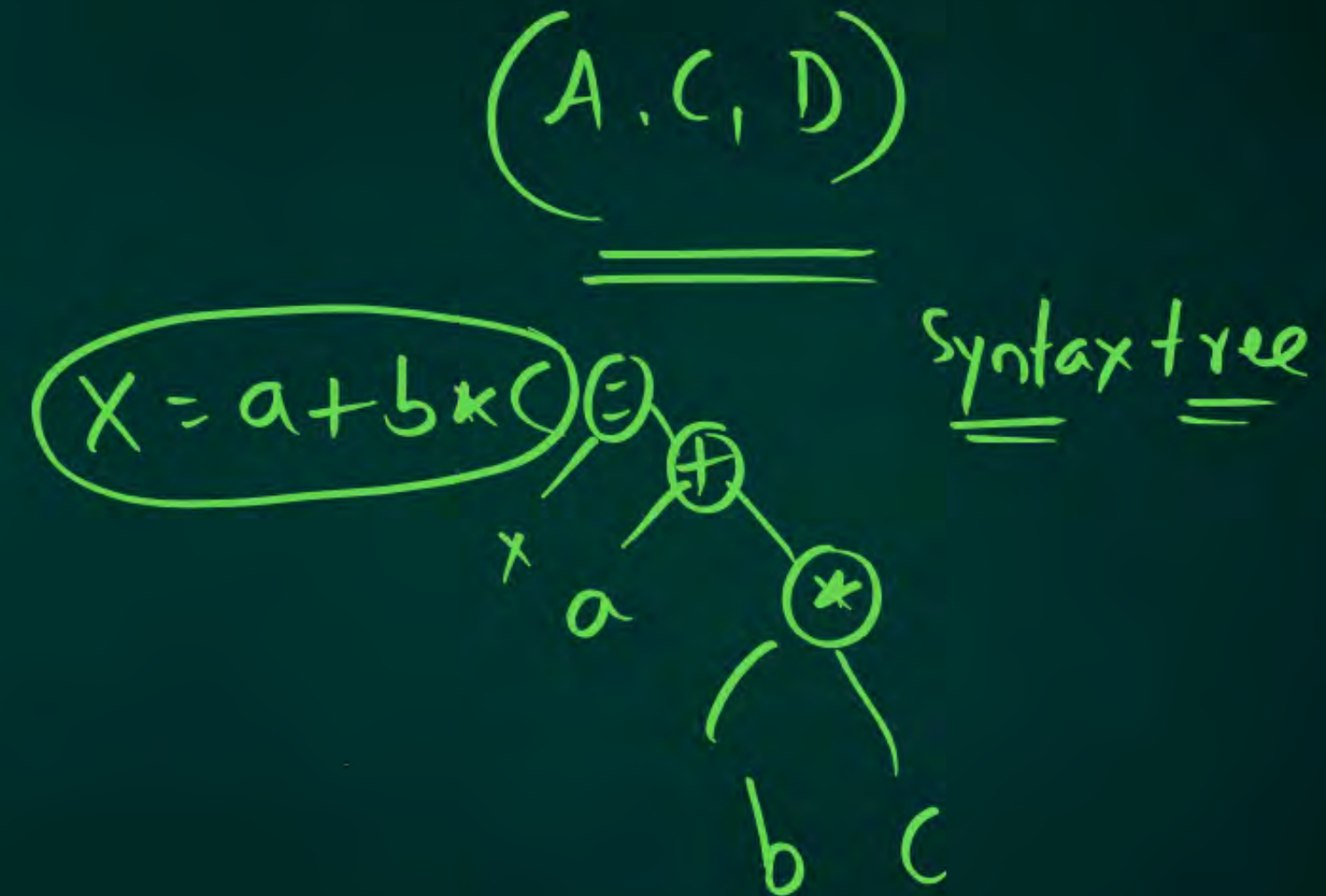
☐ D None of these

(A, B)



#Q. Which of the following is ~~not~~ a type of intermediate code representation used in compilers?

- ☒ A Three-address code
- ☐ B ~~Assembly Language~~
- ☒ C Syntax tree
- ☒ D Postfix notation



Question

#Q. Which of the following is/are token in C language?

- ☒ A Semicolon
- ☒ B Identifier
- ☒ C Keyword
- ☒ D White space

(A, B, C)

Tokens

- ① Identifier
- ② keyword
- ③ Operator
- ④ Constant
- ⑤ Special Symbol

Question

#Q. Which of the following is a job of lexical analysis?

- ☒ **A** Identifying tokens
- ☒ **B** Deleting space and comments
- ☐ **C** Flow control check
- ☐ **D** None of these

(a, b)

Question

#Q. Which of the following compiler has lexical analysis to identify the tokens of the program?

- A** C compiler
- B** C++ compiler
- C** Java compiler
- D** Operating system

Question

#Q. Consider the following grammar :

$S \rightarrow BA$

$A \rightarrow BA/\epsilon$

$B \rightarrow DC$

$C \rightarrow *DCB/\epsilon$

$D \rightarrow (S) /id$

FIRST (S) will be same as that of

☒ A FIRST (B)

☐ B $FIRST(A) = \{ \epsilon, FIRST(B) \}$

☐ C $FIRST(C) = \{ *, \epsilon \}$

☒ D FIRST (D)

$$\underline{\underline{FIRST(S)}} = \underline{\underline{FIRST(B)}} = \boxed{FIRST(D)} = \{ (, id \}$$

(A, D)

Question

#Q. Consider the following grammar G:

$$S \rightarrow \text{epsilon} \mid a = \boxed{S \rightarrow \epsilon \mid a}$$

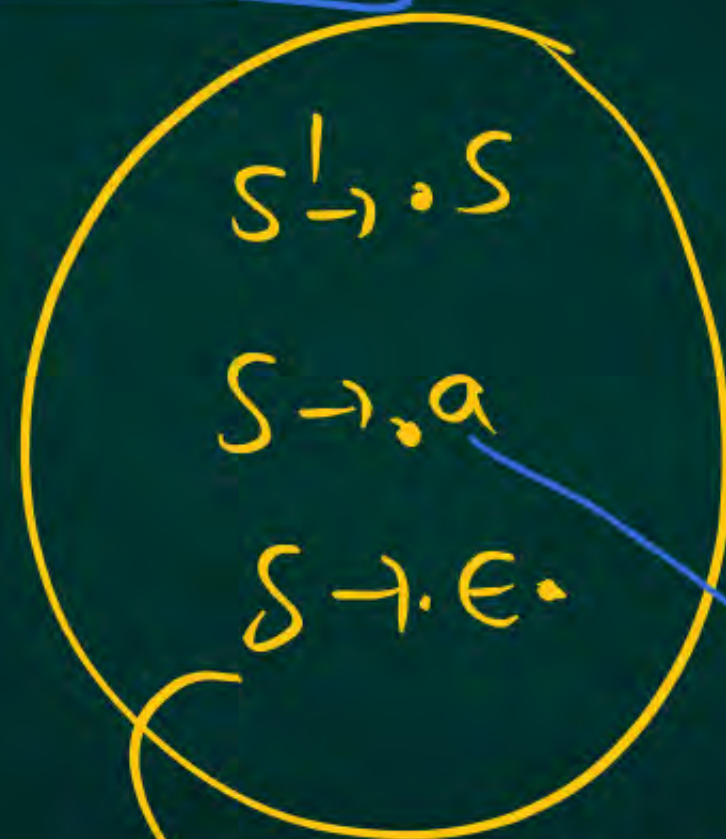
G is _____

☒ A LL(1)

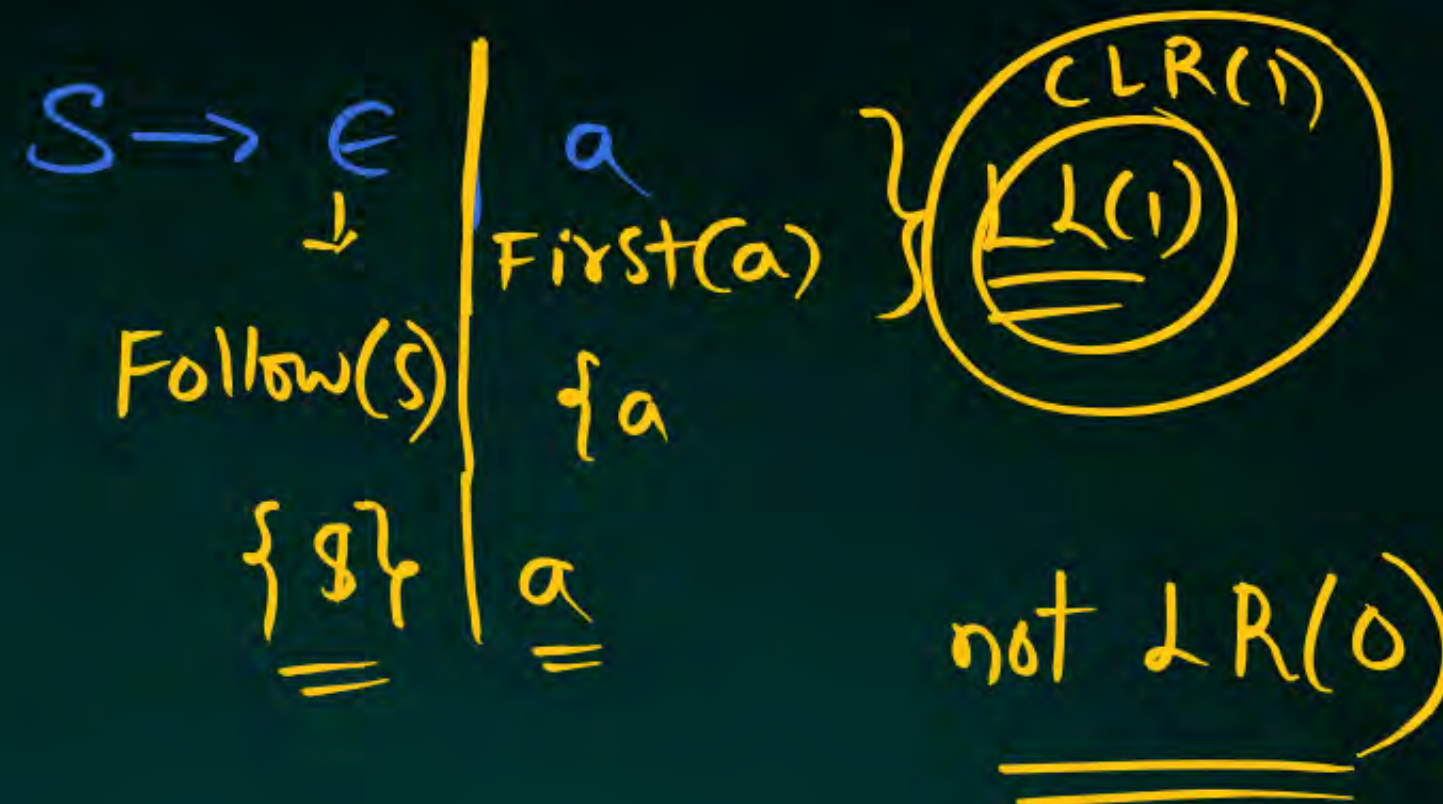
☒ B LR(0)

☒ C SLR(1)

☒ D CLR(1)



$\text{Follow}(S) = \{\$ \}$



✓ SLR(1)

no S/R conflict

Question

#Q. Consider the following Grammar G:

G:

$S \rightarrow Aa \mid Sa \mid d$

$A \rightarrow Ab \mid Sb \mid d$

G is not ____.

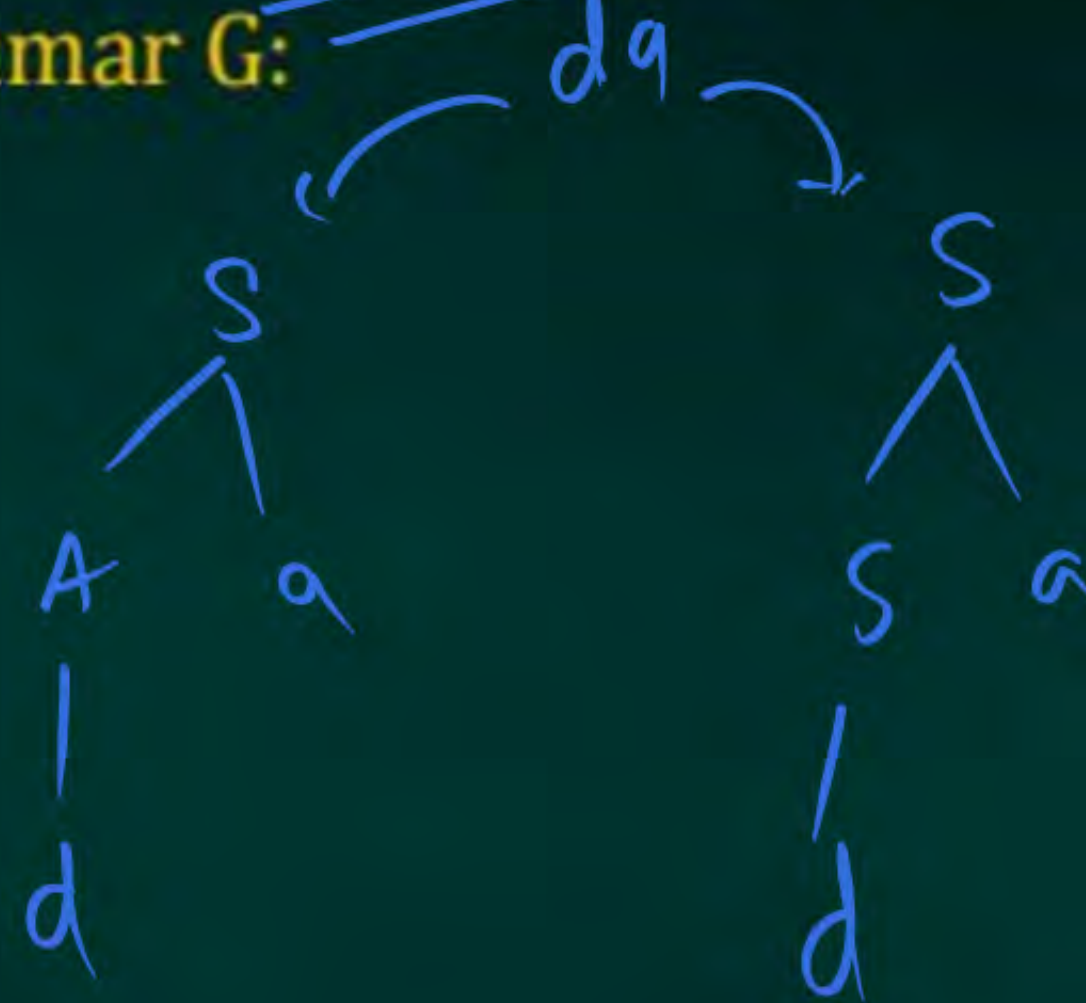
A ~~LL(1)~~

B ~~LR(0)~~

C ~~SLR(1)~~

D ~~CLR(1)~~

Ambiguous grammar



Question

#Q. Which of the following statements (s) is not true?

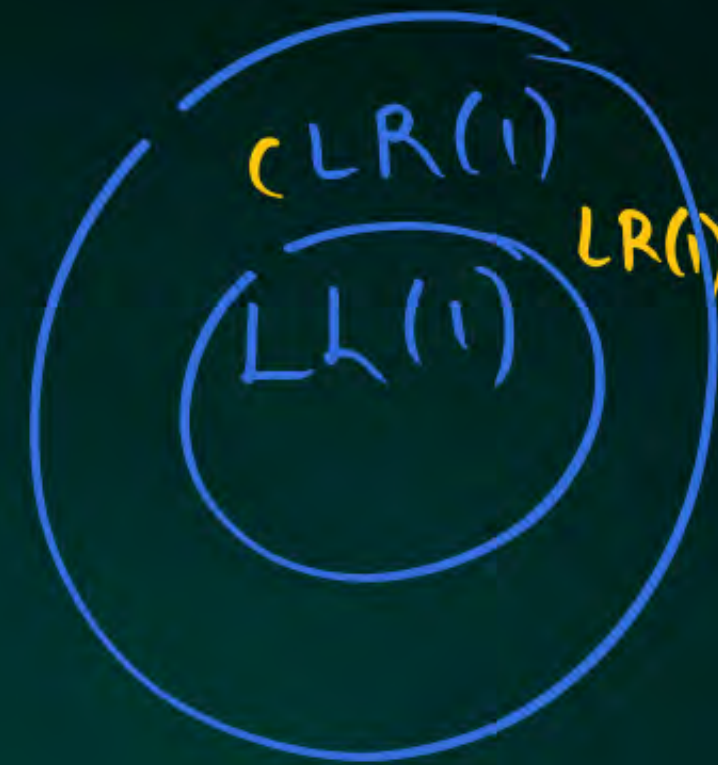
A An LL(1) parser is top down parser

B Ambiguous grammar may be LL(1) *false ✓*

C LL(1) grammar is LR(1)

D LR(1) grammar is LL(1) *false*

(B,D)



Question

#Q. Which of the following grammar is / are not LL(1)?

☒ **A** $S \rightarrow SS \mid \epsilon$ } not LL(1)

(a, b, c)

☒ **B** $S \rightarrow bSc \mid bc$ } not LL(1)

☒ **C** $S \rightarrow Sb \mid c$ } not LL(1)

☐ **D** $S \rightarrow bS \mid c$ } LL(1)

Question

#Q. Which of the following is true?

(A, C)
=

- A** ^{true} If CLR(1) does not contain SR conflict then LALR(1) don't have SR conflict.
- B** ^{false} If CLR(1) does not contain RR conflict then LALR(1) does not contain RR conflict.
- C** ^{true} If CLR (1) does not contain RR conflict then LALR (1) may contain RR conflict.
- D** None

CLR(1) no R/R Conflict

LALR(1) { may have
R/R Conflict

Question

#Q. Which of the following is FALSE?

(A, C)



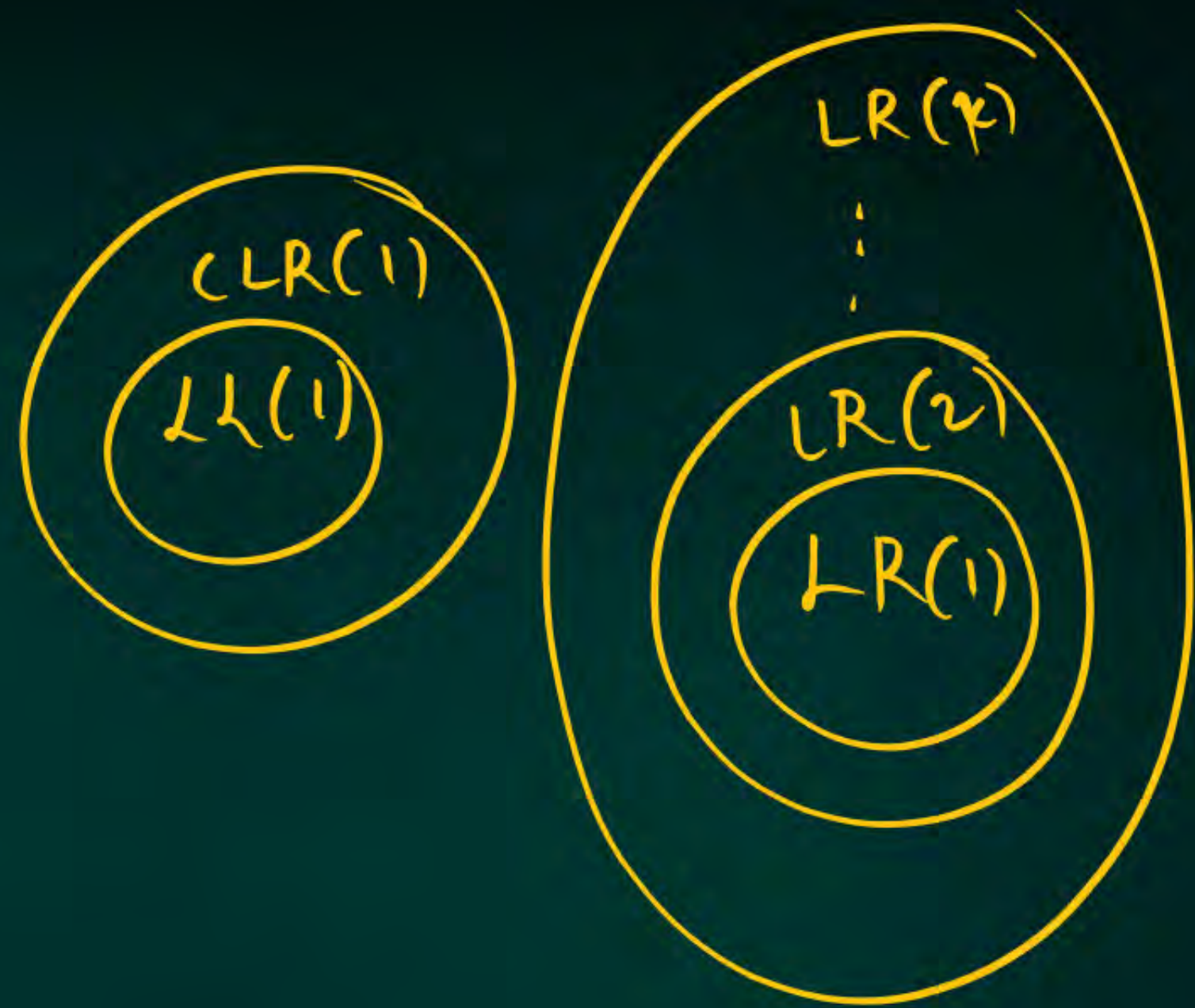
- A** ~~false~~ Every conflict in LR(0) is also a conflict in SLR(1).
- B** ~~true~~ If there is no shift-reduce (SR) conflict in LR(1), then there is no SR conflict in LALR(1).
- C** ~~false~~ If there is no reduce-reduce (RR) conflict in LR(1), then there is no RR conflict in LALR(1).
- D** ~~true~~ Every conflict in LR(1) is also a conflict in SLR(1).

Question

#Q. Every LL(1) CFG is _____

- ☐ A SLR(1) ✗
- ☐ B LALR(1) ✗
- ☒ C LR(1)
- ☒ D LR(2)

(C, D)



Question

#Q. Consider the grammar
 $S \rightarrow aSbS \mid bSaS \mid \epsilon$

Which of the following is/are false?

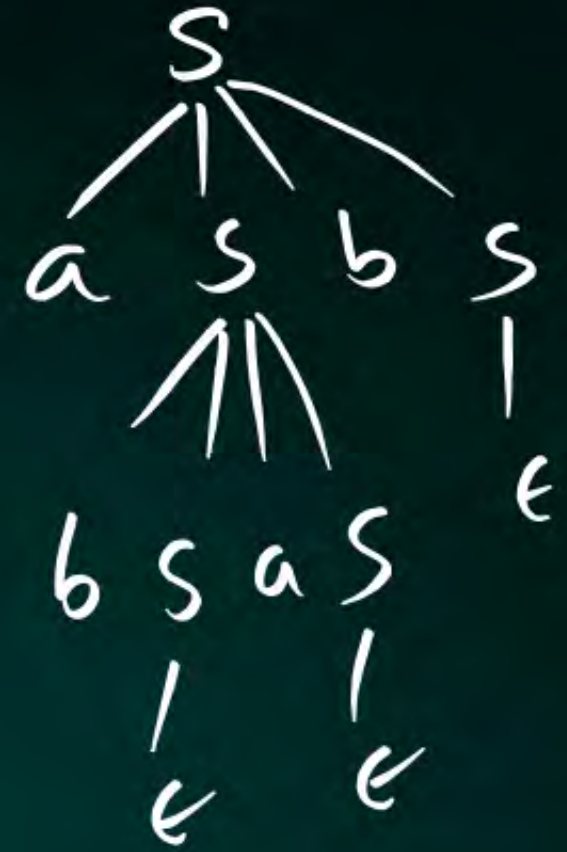
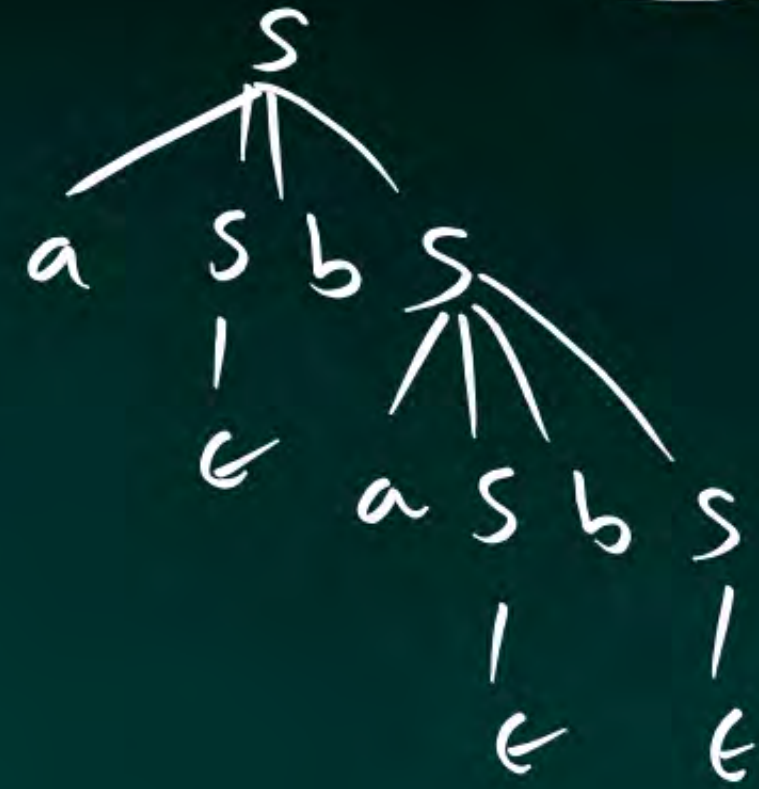
(B, C)

A ^{true} String "abab" will have 2 parse trees

B ^{false} The grammar is ambiguous and string "abab" will have 3 parse trees

C ^{false} The grammar is unambiguous

D ^{true} The grammar is ambiguous.



abab

Question

#Q. Consider the following productions:

$S \rightarrow aABe$

$A \rightarrow Abc | b$

$B \rightarrow d$

(A, B, C)

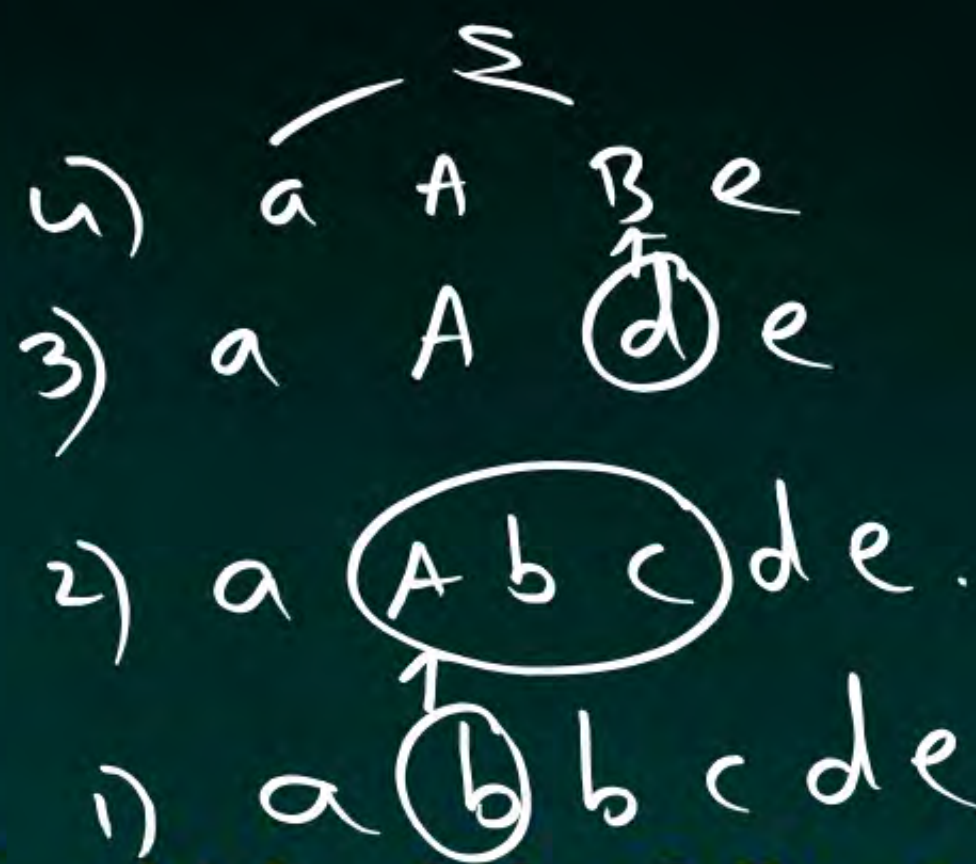
If the sentence "abbcde" is reduced to S, which of the following is/are represent handle (Using bottom up parsing)?

☒ A Abc ✓

☒ B b ✓

☒ C d ✓

☐ D None



Question

#Q. Which of the following is/are TRUE? (A, B, C)

- ☒ **A** ^{true} If CFG has SR conflict in LR(1) parser, then same SR conflict also present in LR(0).
- ☒ **B** ^{true} If CFG has SR conflict in LR(1) parser, then same SR conflict also present in SLR.
- ☒ **C** ^{true} If CFG has SR conflict in LR(1) parser, then same SR conflict also present in LALR.
- ☐ **D** None of these

Question

#Q. Which of the following is correct regarding FIRST & FOLLOW of the given grammar.

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow \underline{F}T'$$

$$T' \rightarrow * \underline{F}T' \mid \epsilon$$

$$F \rightarrow id \mid (E)$$

$$Follow(T) = First(E')$$

$$\{+, \epsilon\}$$

First	Follow
$\{id, (\}$	$\{ \$,) \}$
$E' \{+, \epsilon\}$	$\{ \$,) \}$
$T \{id, (\}$	$\{ +, \$,) \}$
$T' \{*, \epsilon\}$	$\{ +, \$,) \}$
$F \{id, (\}$	$\{ *, +, \$,) \}$

A ~~✗~~ FIRST (T) = {id, (}
FOLLOW(T') = {+, \$,

B ~~✓~~ FIRST (E) = {id, (}
FOLLOW(E') = { \$,) }

C ~~✓~~ FIRST (E) = {id, (}
FOLLOW(F) = {+, \$,), *}

D ~~✗~~ FIRST (T') = {*, ϵ }
FOLLOW (E) = { \$, *,) }

[MSQ]



(A, D)

#Q. Which of the following statement(s) is/are TRUE while computing First and Follow during top down parsing by a compiler?

abc\$

$Follow(S) = \{ \$ \}$

$First(A) = \{ \epsilon \}$

- ☒ A For a production $A \rightarrow \epsilon$, ϵ will be added to $First(A)$.
- ☒ B ~~false~~ If there is any input right end marker, it will be added to $First(S)$, where S is the start symbol.
- ☒ C ~~false~~ For a production $A \rightarrow \epsilon$, ϵ will be added to $Follow(A)$.
- ☒ D If there is any input right end marker, it will be added to $Follow(S)$, where S is the start symbol.



Thank
You

